

Transplant Volumes and Racial Disparities: A Data Analysis of U.S. Organ Allocation

(2018 to 2025)

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Introduction:

Organ transplantation is a critical, life-saving procedure for individuals with end-stage organ failure, yet persistent disparities continue to shape who receives a transplant and who does not. Each year, hundreds of thousands of people are added to the transplant waiting list. In fact, a new name is added every eight minutes — a change from every eleven minutes in 2016. By mid-2025, more than 103,000 men, women, and children were waiting for a transplant. Although more than 48,000 transplants were performed in 2024, the demand continues to outpace the supply, and on average, 13 people die each day while waiting for an organ.

Becoming a donor requires meeting multiple criterias, including age, medical history, and family history, and even after meeting those requirements, further testing is needed to ensure the recipient's body will accept the organ rather than reject it. In some cases, race and ethnicity are also considered during matching, as genetic and immunologic factors can influence compatibility. While sharing the same ethnicity is not required for organ donor–recipient matching, matches between people of different racial or ethnic backgrounds are actually quite common. However, having a larger and more diverse donor registry improves the chances of finding a suitable match for everyone on the waiting list. Because the genetic markers used for matching are inherited, individuals with less common genetic markers may still have a better chance of finding a match among donors who share a similar racial or ethnic background. Prior research has documented differences in transplant access by race, ethnicity, age, and donor type, but many of these disparities remain poorly understood when examined over time and across populations. This study uses data from 2018 to 2025 to examine changes in transplant volumes, the distribution of living versus deceased donor transplants, and patterns among patients on the waiting list. Through statistical summaries, year-to-year comparisons, and data visualization in R Studio, this analysis aims to highlight inequities and better understand who is waiting, who receives transplants, and how these trends have evolved. Gaining a clearer picture of these dynamics is crucial for improving transplant policy and promoting more equitable access to organ transplantation.

Methods:

This study used publicly available transplant datasets covering the years 2018 through 2025, which included information on transplant volumes by race and ethnicity, donor type (living or deceased), patient age groups, and the size of the waiting list. Data cleaning and preparation were completed in R Studio, where the raw tables were transformed into tidy data frames for easier analysis. Descriptive statistics were calculated to capture averages, maximums, minimums, and standard deviations of transplant counts across different racial and ethnic groups. In addition, year-over-year percentage changes were calculated using functions such as `mutate` to better understand shifts over time.

To help visualize patterns and trends, the study relied on `ggplot2`, a code used to graph datasets in R Studio, to build line plots, faceted plots, and summary charts that showed transplant activity by year, by donor type, and across different racial groups. Waiting list data was also summarized and compared against actual transplant rates to highlight any mismatches between supply and demand. The analysis looked at how social or demographic factors might contribute to persistent disparities in transplant access. Before generating the final summaries and visualizations, the data was carefully reviewed to ensure accuracy and consistency. Together, these methods aimed to build a clearer picture of how transplant volumes have changed, who is waiting, who is receiving transplants, and what gaps remain across the system in the United States.

Results:

Between 2018 and 2024, transplant volumes increased steadily across all racial and ethnic groups, reflecting gradual improvements in transplant activity over time. This upward trend remained consistent across most years, signaling growing access to organ transplantation and possibly improvements in donor availability and medical infrastructure. However, in 2025, there was a sudden and significant drop in transplant numbers across all groups. The cause of this decline is not entirely clear. One likely explanation is that the data for 2025 may be incomplete, as the year had not fully concluded at the time of analysis. In large-scale health datasets, especially those involving ongoing clinical reporting, it is common for recent years to

show temporary dips due to delayed submissions or late updates. It's also possible that changes in reporting systems, data availability, or healthcare disruptions played a role.

A smaller but noticeable decrease was also observed in 2020, which likely reflects the impact of the COVID-19 pandemic. During that time, hospitals faced widespread strain, elective surgeries were paused in many regions, and transplant procedures were delayed due to safety concerns, limited ICU capacity, and the risk of viral exposure. Despite these setbacks, transplant volumes rebounded in subsequent years. Notably, in 2019, year-over-year growth was especially strong for Black and Hispanic/Latino patients, suggesting some progress in expanding access during that period. Still, the 2025 dip serves as a reminder of how sensitive transplant systems can be to external factors, and how important it is to interpret recent data with caution when the reporting window may still be open.

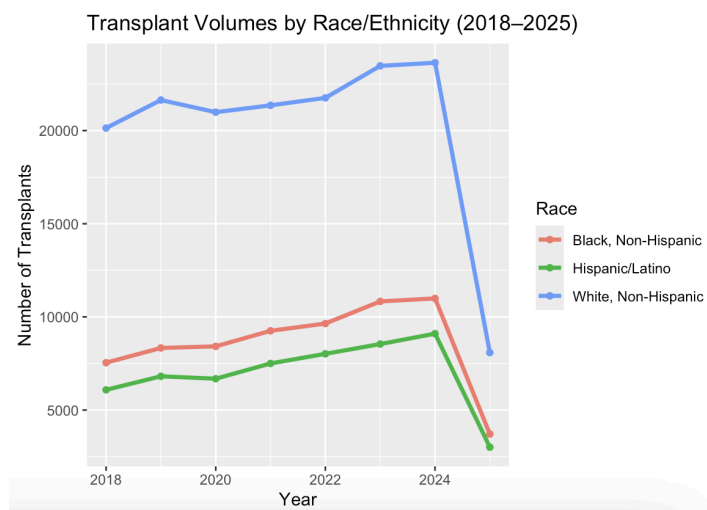


Figure 1. Transplant volumes by race from 2018 to 2025. All groups experienced steady growth in transplant numbers until a noticeable drop in 2025.

Throughout the study period from 2018 to 2025, a clear and consistent disparity emerged in transplant volumes across racial and ethnic groups. White, Non-Hispanic patients received the highest number of transplants each year, averaging approximately 20,133 transplants annually. In

comparison, Black, Non-Hispanic patients averaged around 8,590 transplants per year, while Hispanic/Latino patients received even fewer, with an average of 6,968 annually. This gap remained steady across all observed years, underscoring a persistent imbalance in transplant access. Additionally, White patients exhibited the highest variability in transplant numbers year to year, with a standard deviation of 5,011. This higher variation could indicate greater responsiveness to changing circumstances, more flexible access to healthcare resources, or simply more capacity within their donor networks. In contrast, transplant volumes for Black and Hispanic/Latino patients remained not only lower overall but also less variable, suggesting more rigid and consistent barriers to access. These patterns are not only statistically significant but socially meaningful, they point toward longstanding structural and systemic factors that continue to limit equitable access to life-saving organ transplants for communities of color.

A tibble: 3 × 5

	Race	Average	Max	Min	SD
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Black, Non-Hispanic	8590.	10992	3710	2310.
2	Hispanic/Latino	6968	9100	3008	1889.
3	White, Non-Hispanic	20133.	23640	8083	5011.

Figure 2. Statistical summary of transplant counts by race and ethnicity, showing the average, maximum, minimum, and variability over the study period.

Across all racial and ethnic groups, the majority of organ transplants were performed using deceased donor organs. This reliance on deceased donors was most evident among White, Non-Hispanic patients, who received an average of approximately 16,228 deceased donor transplants per year, nearly double the amount received by Black and Hispanic/Latino patients. Living donor transplants, while far fewer in number overall, followed a similar pattern of disparity. On average, White, Non-Hispanic patients received 3,905 living donor transplants each year, compared to just 719 for Black, Non-Hispanic patients and 1,040 for Hispanic/Latino patients. These differences are not just numbers, they reflect deeper structural and social

challenges. For example, lower living donor rates among Black and Hispanic populations may be tied to socioeconomic barriers, reduced access to donor networks, limited healthcare education, or mistrust in the medical system. Additionally, cultural perceptions of organ donation and eligibility issues related to health disparities may further reduce the pool of potential living donors for these communities. This disparity in both deceased and living donor access further widens the inequality in transplant outcomes and raises critical questions about fairness and equity in the organ transplant system.

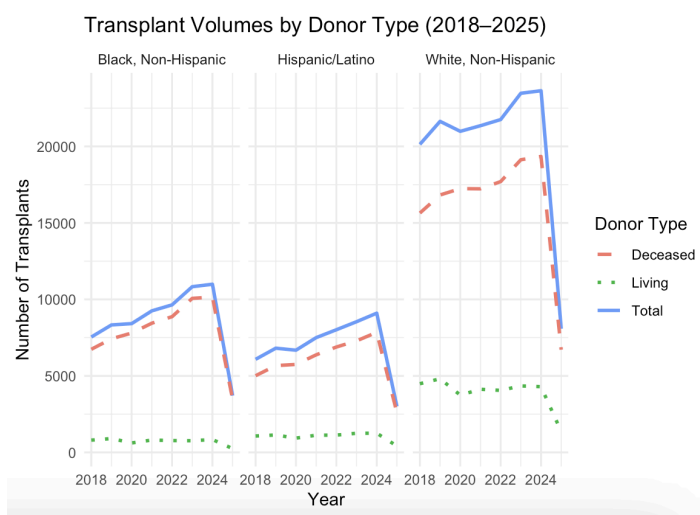


Figure 3. Breakdown of transplants by donor type (deceased vs. living) and race/ethnicity between 2018 and 2025. Deceased donors accounted for the majority of transplants across all groups.

When examining the yearly share of total transplants, it becomes clear that White, Non-Hispanic patients consistently accounted for the largest portion. Despite Black and Hispanic/Latino patients making up a significant percentage of individuals on the waiting list, their share of transplants remained noticeably lower across all years. This imbalance is not just a reflection of population differences — it points to deeper, systemic issues in how organs are allocated and who ultimately receives them. While the need exists across all communities, the actual access to transplant opportunities is unevenly distributed. The fact that White patients continue to receive the majority of transplants year after year, while other groups remain underrepresented in transplant totals, suggests that structural factors — such as socioeconomic

status, access to healthcare, referral rates, and availability of living donors — may be influencing outcomes. This ongoing disparity emphasizes the importance of further investigating the processes that contribute to unequal transplant access, so that more equitable solutions can be implemented.

Race	Year	Total_Transplants	Deceased_Transplants	Living_Transplants	Change_Percent
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 White, Non-Hispanic	2018	20135	15645	4490	NA
2 Black, Non-Hispanic	2018	7541	6739	802	NA
3 Hispanic/Latino	2018	6084	5012	1072	NA
4 White, Non-Hispanic	2019	21639	16826	4813	7.47
5 Black, Non-Hispanic	2019	8332	7428	904	10.5
6 Hispanic/Latino	2019	6811	5673	1138	11.9
7 White, Non-Hispanic	2020	20987	17242	3745	-3.01
8 Black, Non-Hispanic	2020	8418	7800	618	1.03
9 Hispanic/Latino	2020	6681	5749	932	-1.91
10 White, Non-Hispanic	2021	21354	17228	4126	1.75

Figure 4. Yearly share of total transplants by race. White, Non-Hispanic patients consistently received the largest percentage of transplants each year.

As of September 2024, the majority of individuals on the national organ transplant waiting list were adults between the ages of 50 and 64, totaling 43,829 people. This age group represented the largest segment in need of transplants, followed closely by those aged 65 and older (26,460), which signals that the demand for organ transplants grows as people age and develop chronic conditions. Interestingly, even though younger populations also face life-threatening conditions that require transplants, the number of patients under 18 made up a much smaller share of the list, with fewer than 1,000 children waiting. This trend highlights how organ failure tends to affect older populations more prominently, though it remains a pressing concern across all ages.

When breaking down the waiting list by race and ethnicity (based on 2023 data), the largest group was White patients, with 40,476 people awaiting transplants. Black patients were the second-largest group at 28,552, followed by Hispanic/Latino patients at 23,757. While these numbers are consistent with broader population demographics, they also reflect deeper systemic challenges. Even though Black and Hispanic individuals make up a large portion of the waiting list, their chances of receiving a transplant remain disproportionately low. This raises important

questions about whether the system adequately serves all communities or if certain barriers are making it harder for people of color to move from being listed to actually receiving the care that is needed for them to survive.

Discussion:

The results of this study present evidence on the ongoing disparities in organ transplant access across racial and ethnic lines in the United States. From 2018 through 2024, transplant volumes increased for all major groups — a promising trend on the surface. But a deeper look reveals that White, Non-Hispanic patients consistently received far more transplants than Black and Hispanic/Latino patients year after year. The gap wasn't subtle or isolated; it was large, persistent, and repeated annually. Even in years where the waiting list was heavily populated by Black and Hispanic individuals, the actual number of transplants they received remained disproportionately low. This pattern signals a systemic issue that cannot be explained away by population size or health needs alone — it suggests that structural barriers continue to shape who gets access to life-saving care and who doesn't. These disparities are not only unfair; they represent a failure of equity within one of the most critical areas of modern medicine.

Beyond overall volume, the type of donor also played a key role in shaping these disparities. While most transplants came from deceased donors across all groups, living donor transplants revealed an even more significant divide. On average, White, Non-Hispanic patients received nearly four times as many living donor transplants as either Black or Hispanic/Latino patients. This difference is especially concerning because living donor transplants can offer better outcomes and shorter wait times — making them one of the most valuable options in transplant medicine. The low numbers for Black and Hispanic patients don't necessarily reflect a lack of interest or willingness to donate in these communities. Instead, it's likely rooted in systemic challenges — fewer people with health insurance or access to transplant education, more economic hardship that makes donation financially difficult, and a higher likelihood of other health related issues within close family networks that can disqualify potential donors. These disparities reflect broader issues in public health and social infrastructure, and they make it painfully clear that addressing transplant inequities means addressing inequalities far beyond hospital walls.

One of the most noticeable changes in the data came in 2025, when transplant volumes dropped sharply across all racial and ethnic groups. While it's too early to draw definitive conclusions, this sudden decline could be linked to incomplete data collection, delays in reporting, or the fact that the year hadn't yet fully played out when this study was conducted. Still, the dip serves as an important reminder of how sensitive and vulnerable the transplant system can be. When something disrupts the flow — whether it's a pandemic, administrative delay, or reporting issue — the impact can ripple out to affect thousands of people who are depending on that system for survival. That's why it's crucial to look at these trends across multiple years rather than judging the system based on a single year's numbers. The broader patterns we see in this data tell a more powerful story: one of long-standing inequality, persistent gaps, and a healthcare system that continues to fall short in providing equal care for all.

Conclusion:

This study reinforces what many already know but what the data makes undeniable: access to organ transplantation in the U.S. remains unequal. While transplant volumes have increased overall, the distribution of those transplants continues to favor White, Non-Hispanic patients—both in deceased and living donor cases. Meanwhile, Black and Hispanic/Latino patients, despite being a large part of the waiting list, consistently receive fewer transplants.

Addressing this gap isn't just about adding more donors to the system. It's about breaking down the barriers that prevent many patients from even getting a fair shot—barriers tied to education, economics, systemic bias, and unequal access to care. If we want a system that serves everyone, we can't stop at reporting trends; we have to act on what they reveal. That means shifting from awareness to accountability, and from data collection to meaningful, equity-driven change.

References

- Detailed Description of Data* | *organdonor.gov*. (2024, October 17). Organ Donation. Retrieved August 3, 2025, from <https://www.organdonor.gov/learn/organ-donation-statistics/detailed-description#fig1>
- Detailed Description of Data* | *organdonor.gov*. (2024, October 17). Organ Donation. Retrieved August 3, 2025, from <https://www.organdonor.gov/learn/organ-donation-statistics/detailed-description#fig2>
- Organ Transplants and Black/African Americans* | *Office of Minority Health*. (2025, February 13). Office of Minority Health (OMH). Retrieved August 3, 2025, from <https://minorityhealth.hhs.gov/organ-transplants-and-blackafrican-americans>
- Racial and ethnic disparities on the heart transplant waiting list*. (2025, March 15). PubMed. Retrieved August 3, 2025, from <https://pubmed.ncbi.nlm.nih.gov/39814184/>